

JUAN CARLOS CARMENATE

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SUMMARY

Final-year Computer Science student with hands-on experience building Python backends and LLM-powered systems: fine-tuning language models in production, RAG pipelines, and multi-agent infrastructure. Looking for **Backend** or **AI Engineer** roles.

EDUCATION

University of Havana

B.Sc. in Computer Science — final year

Havana, Cuba

Sept. 2023 – Jul. 2027

- Undergraduate thesis on **graph counterfactual explainability (GCE)** (in progress): designing a heuristic node-attribute editing method that flips a graph model's decision to produce counterfactual explanations.

EXPERIENCE

AI Engineer — LLM Fine-tuning

01/2025 – 10/2025 (10 months)

[Private company]

Havana, Cuba

- Fine-tuned a language model to automate the company's internal inventory management, replacing a record-keeping and lookup workflow that until then was manual.
- Built and cleaned the training dataset from internal records, defining the instruction format and the test cases.
- Evaluated the model on real inventory queries and iterated on data and hyperparameters until it reached a quality the team could rely on.

Undergraduate Teaching Assistant — Logic and Discrete Mathematics

2025 – Present

Faculty of Mathematics and Computer Science, University of Havana

Havana, Cuba

- Taught problem-solving sessions in Logic and Discrete Mathematics to first-year Computer Science students.
- Designed and graded exercises and exams, and held office hours on proofs, propositional logic and predicate logic.

PROJECTS

Tech RAG — Information Retrieval System | *Python, FastAPI, ChromaDB, React, TypeScript, Docker* [GitHub](#)

- Built an information retrieval system for the technology domain from scratch: acquisition from 5 APIs (GitHub, Hacker News, Dev.to, StackExchange, arXiv), a custom inverted index with TF-IDF, and a persistent ChromaDB vector store.
- Implemented a **Bayesian Inference Network** retriever (Turtle & Croft, 1991) with noisy-OR and fused it with vector search in a custom RAG pipeline that always answers with citations to the retrieved sources.
- Added query expansion (Rocchio + WordNet), user feedback folded back into ranking, multimodal text→image search with CLIP, and hybrid content-based plus collaborative recommendation.
- Evaluated with Precision@k, Recall@k, MAP, MRR and nDCG against frozen custom qrels, plus RAG faithfulness via LLM-as-judge; 120+ tests and CI on GitHub Actions.

Multiagent Code Generator | *Python, LangGraph, LangChain, Docker, React, WebSockets*

[GitHub](#)

- Orchestrated eight specialized agents with LangGraph to generate a full-stack application (FastAPI + SQLAlchemy / React + Vite) from a natural-language spec.
- Ran the generated code in an isolated Docker sandbox (no network, no access to the Docker socket, CPU and memory limits, non-root user) and closed a self-correction loop with a circuit breaker and an iteration cap.
- Streamed every graph event over WebSocket to a React dashboard showing the active agent, live logs and generated files, replaying full history on reconnect.
- Debugged the system with real end-to-end runs until convergence, fixing agent-routing and prompt defects that stub-based tests could never have caught.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Rust, SQL

AI / ML: LLM fine-tuning, RAG, LangGraph, LangChain, sentence-transformers, ChromaDB, CLIP, Ollama, Anthropic API, NumPy

Backend: FastAPI, SQLAlchemy, SQLite, REST APIs, WebSockets, APScheduler

Frontend: React, Vite, TypeScript

Tools: Docker & Docker Compose, Git, GitHub Actions (CI), Linux, pytest, Vitest

Spoken: Spanish (native), English (B2)